RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Wednesday, 16 September 2026, 11:25 PM
State	Finished
Completed on	Wednesday, 16 September 2026, 11:26 PM
Time taken	1 min 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int find(int a[], int l, int h, int x)
{
    if(l > h)
        return -1;
    int m = (l + h) / 2;
    if(a[m] == x)
        return m;
    if(a[m] < x)
        return find(a, m + 1, h, x);
    return find(a, l, m - 1, x);
}
int main()
{
    int n, x, i, j;
    scanf("%d", &n);
    int a[n];
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
```

```
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for(i = 0; i < n; i++)
    {
        j = find(a, i + 1, n - 1, x - a[i]);
        if(j != -1)
        {
            printf("%d\n%d", a[i], a[j]);
            return 0;
        }
    }
    printf("No");
    return 0;}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course